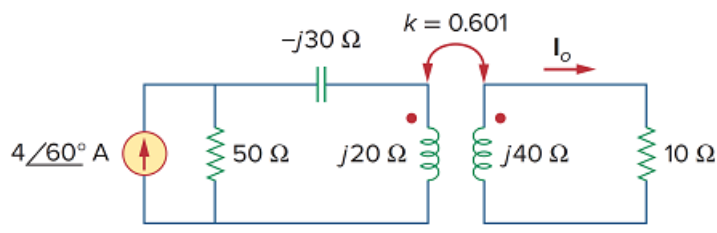


Problem

Find I_o in the circuit of Fig. 13.95. Switch the dot on the winding on the right and calculate I_o again.

Figure 13.95



Step-by-step solution

Step 1 of 7

Refer to Figure 13.95 in the textbook.

Apply source transformation at the primary section.

$$\begin{aligned} V_{eq} &= I_s R \\ &= (4\angle 60^\circ)(50) \\ &= 200\angle 60^\circ \text{ V} \end{aligned}$$

Determine the value of $j\omega M$.

$$\omega M = k\sqrt{\omega L_1 \omega L_2}$$

Substitute 20 for ωL_1 , 40 for ωL_2 , and 0.6 for k in equation .

$$\begin{aligned} \omega M &= k\sqrt{\omega L_1 \omega L_2} \\ \omega M &= 0.6\sqrt{(20)(40)} \\ &= 0.6\sqrt{800} \\ &= 17 \end{aligned}$$

Step 2 of 7

Connect 50Ω in series with $200\angle 60^\circ \text{ V}$ and draw the circuit as shown in Figure 1.

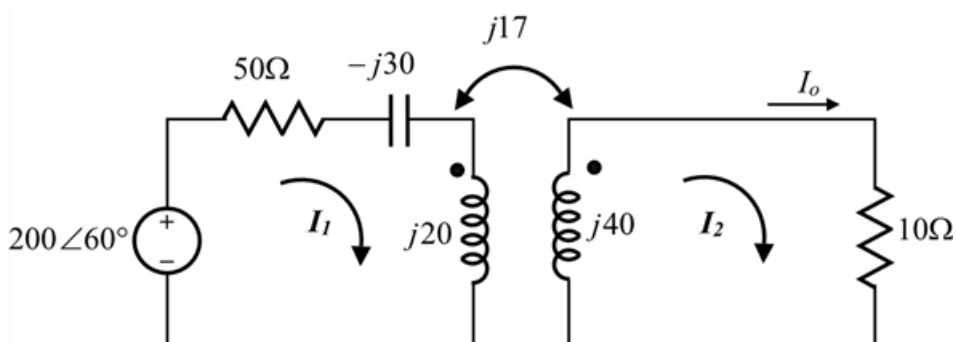


Figure 1

Step 3 of 7

Apply mesh analysis in loop 1.

$$200 \angle 60^\circ = 50I_1 - j30I_1 + j20I_1 - j17I_2$$

$$(50 - j10)I_1 - j17I_2 = 200 \angle 60^\circ \dots\dots (1)$$

Apply mesh analysis in loop 2.

$$10I_2 + j40I_2 - j17I_1 = 0$$

$$-j17I_1 + (10 + j40)I_2 = 0 \dots\dots (2)$$

Step 4 of 7

Write the equation (1) and (2) in matrix form.

$$\begin{bmatrix} 200 \angle 60^\circ \\ 0 \end{bmatrix} = \begin{bmatrix} 50 - j10 & -j17 \\ -j17 & 10 + j40 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \dots\dots (3)$$

Determine the Δ value.

$$\begin{aligned} \Delta &= [(50 - j10)(10 + j40)] - (-j17)^2 \\ &= 500 + j2000 - j100 + 400 + 289 \\ &= 1189 + j1900 \end{aligned}$$

Determine the value of Δ_1 .

Consider the matrix as,

$$\begin{bmatrix} 100 + j173.2 & -j17 \\ 0 & 10 + j40 \end{bmatrix}$$

$$\begin{aligned} \Delta_1 &= [(100 + j173.2)(10 + j40)] - (0)^2 \\ &= -5928 + j5732 \end{aligned}$$

Step 5 of 7

Determine the value of Δ_2 .

Consider the matrix as,

$$\begin{bmatrix} 50 - j10 & 100 + j173.2 \\ -j17 & 0 \end{bmatrix}$$

$$\begin{aligned} \Delta_2 &= j17(100 + j173.2) \\ &= -2944.4 + j1700 \end{aligned}$$

Step 6 of 7

Determine the value of I_2 .

Consider the equation for I_2 .

$$I_2 = \frac{\Delta_2}{\Delta}$$

Substitute $1189 + j1900$ for Δ , and $-2944.47 + j1700$ for Δ_2 in equation.

$$\begin{aligned} I_2 &= \frac{-2944.4 + j1700}{1189 + j1900} \\ &= -0.054 + j1.156 \\ &= 1.517 \angle 92^\circ \text{ A} \end{aligned}$$

From the circuit,

$$\begin{aligned} I_o &= I_2 \\ &= 1.517 \angle 92^\circ \text{ A} \end{aligned}$$

Therefore, the value of current I_o is $\boxed{1.517 \angle 92^\circ \text{ A}}$.

Step 7 of 7

Consider the dot on the winding is changed on right side.

Determine the current, I_o .

Changing the dot on right side changes only the direction of direction of I_o .

$$\begin{aligned} I_o &= 1.517 \angle (180^\circ + 92^\circ) \\ &= 1.517 \angle 272^\circ \\ &= 1.517 \angle -88^\circ \text{ A} \end{aligned}$$

Therefore, the value of current I_o is $\boxed{1.517 \angle -88^\circ \text{ A}}$.